

Ay9 Inferencia:

Calc. Variancia Rao: - Condicion de regularidad $\sigma^2 I_X(\theta) < \infty$
 - Sea T estimador insesgado para $\tau(\theta)$, entonces

$$Var(T) \geq \frac{[\tau'(\theta)]^2}{I_X(\theta)} \quad \forall \theta \in \Theta$$

Rao Blackwell: $X_1, \dots, X_n$ iid $f(x|\theta)$. S estimador suficiente para θ .

Estimador insesgado de $\tau(\theta)$. Sea $T^* = E[T|S]$

$$E(T^*) = \tau(\theta)$$

$$Var(T^*) \leq Var(T) \quad \forall \theta \in \Theta$$

Teorema Lehmann-Scheffé: $X_1, \dots, X_n$ iid $f(x|\theta)$. S estimador suficiente y completo. $T^* = T^*(S)$ y T^* es insesgado para $T^* \rightarrow T^*$ es EIVUM de $\tau(\theta)$

Asintóticamente: - Insesgado: $\lim_{n \rightarrow \infty} E(T_n|\theta) = \tau(\theta)$

$$- ECM \quad \lim_{n \rightarrow \infty} ECM(T_n, \theta) = 0$$

$$- Eficiencia relativa ARE(T_1, T_2, \theta) = \lim_{n \rightarrow \infty} \frac{ECM(T_1, \theta)}{ECM(T_2, \theta)}$$

Método Delta (estimar $\tau(\theta)$ desde $\tau(T)$)

$$\tau(t) \approx \tau(u) + (t-u)\tau'(u)$$

$$Luego: E(\tau(T)) \approx \tau(u) \quad y \quad Var(\tau(T)) = [\tau'(u)]^2 \cdot Var(T)$$

TCL: $X_1, \dots, X_n$ iid $E(X_i) = \mu$ y $Var(X_i) = \sigma^2 < \infty$

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Metodo Delta con TCL

$$\mu = E(S_i), \sigma^2 = Var(S_i)$$

$$T_n = g(S_n) \text{ si } \sqrt{n}(S_n - \mu) \xrightarrow{d} N(0, \sigma^2) \text{ , } g \text{ con deriv. contin.}$$

$$\text{Entonces } \sqrt{n}(T_n - g(\mu)) \xrightarrow{d} N(0, [g'(\mu)]^2 \sigma^2)$$

P1/ Encuentra E y Var de $S(\theta)$, $x_1, \dots, x_n$

$$c) f(x|\theta) = \theta e^{-x\theta} \mathbb{1}_{\{x>0\}}$$

$$f(\bar{x}|\theta) = \theta^n \exp\{-\theta \sum x_i\} \quad \bar{X} = \frac{1}{n} \sum x_i \rightarrow \text{Suficiente y completa}$$

$$E(\bar{X}) = E(x_i) = \int_0^{\infty} x \theta e^{-x\theta} dx$$

EMV

$$= \int_0^{\infty} \theta \cdot -\frac{\partial}{\partial \theta} e^{-x\theta} dx$$

$$= \theta \cdot -\frac{\partial}{\partial \theta} \int_0^{\infty} e^{-x\theta} dx$$

$$= \theta \cdot -\frac{\partial}{\partial \theta} \frac{1}{\theta} \int_0^{\infty} \theta e^{-x\theta} dx$$

$$\theta \cdot -\frac{\partial}{\partial \theta} \frac{1}{\theta} = \theta \cdot \frac{1}{\theta^2} = \frac{1}{\theta} = E(\bar{X})$$

$$Var(\bar{X}) = \frac{1}{n} Var(x_i) = \frac{1}{n\theta^2}$$

$$E(\bar{X}^2) = \frac{1}{n\theta^2} + \frac{1}{\theta^2} = \frac{1+n}{n\theta^2}$$

$$F_X(x) = 1 - e^{-x\sigma}$$

$$X_i \sim \text{Gam}(n, \sigma) = Y$$

$$f_Y(y) = \frac{1}{\Gamma(\alpha)} B^\alpha y^{\alpha-1} e^{-yB}$$

$$= \frac{1}{\Gamma(n)} B^n y^{n-1} e^{-y\sigma n}$$

$$Z = 1/Y \rightarrow Y = 1/Z$$

$$\frac{X_i}{n} = y_i \sim \exp(n\sigma)$$

$$X_i = n y_i$$

$$dx_i = n dy_i$$

$$n p(1-p)^{n-1}$$

$$Y \sim \text{Gam}(n, \sigma)$$

$$\mathbb{E}\left(\frac{1}{Y}\right) = n \mathbb{E}\left(\frac{1}{Y}\right) = n \int_0^\infty \frac{1}{y} \frac{1}{\Gamma(n)} \sigma^n y^{n-1} e^{-y\sigma} dy$$

$$= \frac{n}{\Gamma(n)} \frac{\Gamma(n-1)\sigma}{\Gamma(n-1)} \int_0^\infty \frac{1}{\Gamma(n-1)} \sigma^{n-1} y^{n-2} e^{-y\sigma} dy$$

$$= \frac{n}{\Gamma(n)} \frac{\Gamma(n-1)\sigma}{\Gamma(n-1)} \cdot 1$$

$$\mathbb{E}\left(\frac{1}{Y}\right) = \frac{n}{n-1} \sigma$$

$$\mathbb{E}\left(\frac{1}{Y}\right) = \frac{\sigma}{n-1}$$

$$\mathbb{E}\left(\frac{n-1}{Y}\right) = \boxed{\sigma}$$

$$T^* = \frac{n-1}{S} \quad \begin{array}{l} \text{mesurado} \\ \text{après} \\ \text{rot de } S \end{array}$$

$$T^* \text{ EIV de } S(\sigma) = \sigma.$$

P1.5 $P_X(x|\theta) = \binom{K}{x} \theta^x (1-\theta)^{K-x}$

$$g(\theta) = P_r(X=1) = K\theta(1-\theta)^{K-1}$$

$$\mathbb{I}(X_1=1) \text{ es insesgado} \Rightarrow \bar{1} = 1 \text{ si } X_1=1$$

$$E(\mathbb{I}(X_1=1)) = P_r(X_1=1) = g(\theta)$$

$S = X_1 + \dots + X_n$ es suficiente y completa

$$S \sim B_{in}(nK, \theta)$$

Familia exponencial

No Blackall

$E(\mathbb{I}(X_1=1) | S)$ es e.v.un de $g(\theta)$

$$P_r(X_1=1 | S) = \frac{P_r(X_1=1, X_1 + \dots + X_n = S)}{P_r(X_1 + \dots + X_n = S)} = \frac{P_r(X_1=1) P_r(X_2 + \dots + X_n = S-1)}{P_r(S=S)}$$

$$\frac{\binom{(n-1)K - (S-1) + K - 1}{K-1} \theta^{K-1} (1-\theta)^{(n-1)K - (S-1)} \cdot \binom{(n-1)K}{S-1} \theta^{S-1} (1-\theta)^{(n-1)K - (S-1)}}{\binom{nK}{S} \theta^S (1-\theta)^{nK-S}}$$

$$= \frac{\binom{(n-1)K}{S-1} K}{\binom{nK}{S}}$$

P2 | $p \in (0, 1)$ i.i.d. $X_1, \dots, X_n$ and $T_n(\text{sum})$ $V_n(X) < \infty$

$$\sqrt{n}(\bar{X} - E(X)) \rightarrow N(0, V_n(X))$$

E_n number car

$$E(Y_n) = E(X_1) = 1/p$$

$$V_n(X_1) = \frac{1-p}{p^2}$$

$$\downarrow$$

$$V_n(Y_n) = \frac{1}{n} V_n(X_1) = \frac{1-p}{np^2}$$

$$\sqrt{n}(Y_n - 1/p) \xrightarrow{d} N(0, (1-p)/p^2)$$

b) $p(1-p) \rightarrow V_{n \rightarrow \infty}$

$$Y_n(1-Y_n)$$

$$g(p) = p(1-p)$$

$$g'(p) = 1-2p$$

$$g(Y_n) \approx g(p) + g'(p)(Y_n - p)$$

$$E(g(Y_n)) \approx g(p)$$

$$V_n(g(Y_n)) \approx [g'(p)]^2 V_n(Y_n)$$

Levon

$$\sqrt{n}(Y_n(1-Y_n) - p(1-p)) \rightarrow N(0, (1-2p)^2 \frac{1-p}{p^2})$$

P3) $X_1, \dots, X_n \sim^{\text{iid}} P(\lambda)$

$E(X) = \lambda$

$V(X) = \lambda$

$P_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}$

$\sqrt{n}(\bar{X} - \lambda) \xrightarrow{L} N(0, \lambda)$ T C L

$g(\lambda) = e^{-\lambda}$

$g'(\lambda) = -e^{-\lambda}$

$g(\bar{X}) = g(\lambda) + g'(\lambda)(\bar{X} - \lambda)$

$\sqrt{n}(e^{-\bar{X}} - e^{-\lambda}) \rightarrow N(0, [g'(\lambda)]^2 V(X))$

$\sqrt{n}(e^{-\bar{X}} - e^{-\lambda}) \rightarrow N(0, e^{-2\lambda} \lambda)$

$\ell_{P_X} = x \ln(\lambda) - \lambda + c$

$\ell L = \sum x_i \ln(\lambda) - n\lambda$

$\frac{\partial \ell L}{\partial \lambda} = \frac{\sum x_i}{\lambda} - n = 0$

$\bar{X} \stackrel{\text{EMV}}{=}$

$\frac{\partial^2 \ell L}{\partial \lambda^2} = -\frac{\sum x_i}{\lambda^2} < 0$

EMV de

$\text{Exp}(-\lambda)$

e)

$\exp(-\hat{\lambda})$

$\exp(-\bar{X})$

P4 $X_1 \dots X_n \text{ i.i.d } U_{n,F}(0, \sigma)$

$$F_X(x) = \frac{x}{\sigma} \rightarrow 1 - \frac{x}{\sigma} = \frac{\sigma - x}{\sigma}$$

$$\hat{\sigma}_1 = 2\bar{X}$$

$$E(\hat{\sigma}_1) = 2E(X_1)$$

$$= 2 \frac{\sigma}{2} = \sigma \text{ unbiased}$$

$$\hat{\sigma}_3 = X_{(n)} \sim f = n f_X F_X^{n-1} \\ = n \frac{1}{\sigma^n} \cdot x^{n-1}$$

$$E(\hat{\sigma}_3) = \int_0^\sigma \frac{n x}{\sigma^n} x^{n-1} dx \\ = \frac{n}{\sigma^n} \frac{\sigma^{n+1}}{n+1}$$

$$= \frac{n\sigma}{n+1}$$

$$= \frac{\sigma}{1 + 1/n} \xrightarrow{n \rightarrow \infty} \sigma$$

$$\hat{\sigma}_2 = (n+1) X_{(1)}$$

$$X_{(1)} \sim f_{X_{(1)}}(x) = n F_X(x) [1 - F_X(x)]^{n-1}$$

$$= n \frac{1}{\sigma} \cdot \frac{(\sigma - x)^{n-1}}{\sigma^{n-1}} = n \frac{(\sigma - x)^{n-1}}{\sigma^n}$$

$$E(\hat{\sigma}_2) = (n+1)E(X_{(1)})$$

$$= \int_0^\sigma x \frac{n (\sigma - x)^{n-1}}{\sigma^n} dx$$

$$= \frac{n}{\sigma^n} \int_0^\sigma x (\sigma - x)^{n-1} dx$$

$$= \frac{n}{\sigma^n} \int_0^\sigma (\sigma - y) y^{n-1} dy$$

$$= \frac{n}{\sigma^n} \left[\frac{\sigma^{n+1}}{n} - \frac{\sigma^{n+1}}{n+1} \right]$$

$$= \frac{n}{\sigma^n} \frac{\sigma^{n+1}}{n(n+1)} = \frac{\sigma}{n+1}$$

$$E(\hat{\sigma}_2) = n+1 \cdot \frac{\sigma}{n+1} = \sigma$$

p5) $X_1, \dots, X_n$ iid $F_{\theta}(x|\theta) = [G(x)]^\theta$ $\theta > 0$
 $\downarrow$ FDA Concord

$$f(x|\theta) = \theta g(x) [G(x)]^{\theta-1}$$

$$L = \theta^n \prod_{i=1}^n g(x_i) [G(x_i)]^{\theta-1}$$

$$\ln L = n \ln \theta + \sum \ln g(x_i) + (\theta-1) \sum \ln G(x_i)$$

$$\frac{\partial L}{\partial \theta} = \frac{n}{\theta} + \sum \ln [G(x_i)] = 0$$

$$\left(\frac{1}{\theta}\right) = - \frac{\sum \ln [G(x_i)]}{n} = - \frac{\sum \ln ([G(x_i)]^\theta)^{1/\theta}}{n} = - \frac{\sum \ln (F(x_i)^{1/\theta})}{n}$$

$$X \sim F \rightarrow F(X) \sim U_{\text{unif}}(0,1)$$

$$F_{F(X)}(x) = P(F(X) \leq x) = P(X \leq F^{-1}(x)) = F(F^{-1}(x)) = x$$

$$F(X_i) \sim U, \quad -\ln(U^{1/\theta}) = z$$

$$U^{1/\theta} = e^{-z}$$

$$f_z = 1/\theta f_U(x)$$

$$f_z = \theta e^{\theta z} \sim \exp(\theta)$$

$$U = e^{-\theta z}$$

$$dU = -\theta e^{-\theta z} dz$$

$$\left(\frac{1}{\theta}\right) = - \frac{\sum \ln (F(x_i)^{1/\theta})}{n} = \frac{\sum z_i}{n}, \quad z_i \sim \exp(\theta)$$

$$E\left(\frac{1}{\theta}\right) = E(z_i) = 1/\theta \quad \text{result}$$

$$V_a\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \frac{1}{n} V_a(z_1) = \frac{1}{n\sigma^2}$$

Cela donne aussi

$$\begin{aligned} I(\sigma) &= E\left[\left(\frac{\partial}{\partial \sigma} \ell(\sigma)\right)^2\right] = -E\left[\left(\frac{\partial^2}{\partial \sigma^2} \ell(\sigma)\right)\right] \\ &= -E\left[\left(-\frac{n}{\sigma^2}\right)\right] = \frac{n}{\sigma^2} \end{aligned}$$

$$\begin{aligned} h(\sigma) &= \frac{1}{\sigma} \\ h'(\sigma) &= -\frac{1}{\sigma^2} \end{aligned}$$

Si l'on insère la cta de crm rai

$$\frac{[h'(\sigma)]^2}{I_x(\sigma)} = \frac{1/\sigma^4}{n/\sigma^2} = \frac{1}{n\sigma^2} \quad \text{Alors la cta}$$